

Michele Bosio

MILANO - BERGAMO

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Summary

Currently wrapping up my Master Thesis focused on an **XR Multiplayer Application**. Passionate about immersive technologies, creative problem-solving, and building systems that scale. Background in **Software Development**, **Machine Learning**, and **Data Science**, with a evergreen interest in the **Videogames** industry. Always eager to learn new tools and technologies when the challenge demands it. **Looking for** opportunities to contribute to innovative teams solving complex problems.

Education

MsC in Software Engineering and Computer Science

POLIMI - POLITECNICO DI MILANO

Milan, Italy

Sep 2022 – Jul 2026

- **Relevant courses:** Artificial Neural Networks and Deep Learning, Streaming Data Analytics, Machine Learning, Advanced Algorithms and Parallel Programming, Principles of Programming Languages, Advanced Computer Architectures, Distributed Systems, Robotics.

Bachelor Degree in Software Engineering and Computer Science

POLIMI - POLITECNICO DI MILANO

Milan, Italy

Sep 2018 – Sep 2022

Projects

Master Thesis — Mixed Reality Physics Interaction System (Unity, XR)

Mar 2025 – Jul 2026

[UNITY, C#, XR]

- Master's Thesis in partnership with the MiBTec Research Laboratory.
- Developed in **Unity (C#)** with XR Interaction Toolkit and hand-tracking input for natural gestures.
- Designed a **client-authoritative multiplayer architecture** and custom raycast/UI systems for precise control, reducing jittering and physics desync by ~ 40%.
- Planned user experience evaluation focusing on immersion, accuracy, and interaction fidelity.

Robotics Course - ROS Project

Mar 2024 – Jun 2024

[C++, ROS, SLAM]

- Collaborated in a 3-person international team to develop a **ROS** package integrating data from 3 sensors (GPS, wheel odometry, and LiDAR) for real-time RViz visualization.
- Built a dynamic reconfiguration system for real-time frame switching between GPS and wheel odometry.
- Implemented **SLAM-based mapping**, localization, and pathfinding algorithms for autonomous navigation.
- Focused on sensor fusion, environment modeling, and navigation within the ROS framework.

Computer Graphics Course - Pacman 3D

Nov 2023 – Feb 2024

[C++, VULKAN, BLENDER]

- **Led technical development** of a 3D PacMan-style game in a 3-student group, coordinating coding tasks and milestones.
- Implemented 3D rendering and scene management from scratch using **C++** and **Vulkan SDKs**.
- Created 10+ basic 3D models in Blender and integrated audio systems to enhance interactivity.
- Focused on **team collaboration**, code modularity, and **performance optimization**, achieving stable 120FPS with manual Vulkan optimization.

Design and Implementation of Mobile Applications Course - PokeDima

Feb 2024

[FLUTTER, DART, FIREBASE, ANDROID, IOS]

- Full-stack mobile app using **Flutter (Dart)** with **Firestore** backend DB and responsive UI for multiple screen sizes, completed independently.
- Implemented **real-time user interaction** and **data synchronization** for online card trading.
- Designed authentication, profiles, and card-exchange logic ensuring efficient data handling.
- Designed and executed structured testing sessions, playtested with 5 users and 50+ cards to ensure stability and error-free interaction.

Videogames Design and Programming Course - Reclaim Humanity

Nov 2023 – Feb 2024

[UNITY, C#]

- **Led a 5-member multidisciplinary team** to build a multi-level RPG emphasizing exploration and adventure.
- Created custom graphics, sound, and environmental assets, and implemented game logic and player controls.
- Ensured smooth transitions and responsive gameplay across multiple areas.

Artificial Neural Networks and Deep Learning Course - ANN Challenge

Sep 2023 – Dec 2023

[CNNs, DATA PROCESSING, FINE TUNING, TRANSFER LEARNING]

- Two-phase competitive project classifying 10K+ healthy vs. diseased plant leaves images using **neural networks**.
- Developed from **CNN to deep models**, with **automated data-processing** pipeline to remove irrelevant images.
- Collaborated with peers to optimize performance via experimentation and feature engineering, achieving an improve of ~ 12% in accuracy.

Personal Project - Python Game Automation Agents

2022 – 2023

[PYTHON, SELENIUM, MULTITHREADING, MULTIPROCESSING, REINFORCEMENT LEARNING]

- Developed **high-performance** automation **python** scripts for web-based monthly challenges.
- Implemented **multithreading** and **multiprocessing** algorithms, achieving an improve in performances of ~ 500%.
- Wordle solver with **operational research optimization** achieved **top-5** placements
- Tetris-like **Reinforcement Learning agent** reached frequent **top-15** finishes.
- Ensured **maximum reliability** and uptime: scripts ran a full week on a server, monitored with custom CLI, with > 99.99% uptime .